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Three-dimensional divergent hydroheteroarylation of internal alkynes with indolizines

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Divergent synthesis approaches generate all isomers of a molecule, but typically focus on single-dimensional selectivity control. The more challenging two-dimensional divergent synthesis dictating two types of selectivity to achieve all four isomers has recently received some attention. As such, three-dimensional divergent synthesis through dictation of three types of selectivity to prepare eight isomers or analogues is very challenging. Here we report a catalyst-controlled three-dimensional divergent protocol for the synthesis of eight types of diaryl allyl skeleton (through combinations of *E*/*Z* and *R*/*S* stereochemistry and 1,1-/1,3-substitution patterns) from the reaction of alkynes and heteroarenes. Good geometric selectivity, regioselectivity and enantioselectivity are observed for the hydroheteroarylation process of internal alkynes with indolizines via Pd and Rh catalysis. Experimental data and density functional theory calculations reveal that the Pd and Rh catalysts have differing mechanistic pathways for the hydroarylation of the internal alkynes and corresponding allene intermediates.

Differing stereochemistry and isomer structure can lead to very different pharmacologic effects[1,2]. The development of a transformation method that enables access to all isomers (enantio-, diastereo-, regio- and geometric isomers) of a target molecule from the same or similar substrates via divergent synthesis has therefore garnered widespread interest[3–14]. At present, most divergent studies focus on one-dimensional control, such as only enantiocontrol, to selectively access both enantiomers of a chiral motif[8–11] (Fig. 1a). Due to the benefits of diversity and three-dimensionality in molecule structure for bioactivity profiles[15,16], skeletons are designed to contain different types of stereoselectivity, which leads to the generation of multiple possible isomers. The synthetic difficulty in readily accessing all corresponding isomers has thus become considerable. For example, when a molecule requires simultaneous control over two factors, such as regioselectivity and geometric configuration, methods for the convenient divergent synthesis of all four possible isomers remain limited. In this domain, stereodivergent synthesis[3,4,17–21] via diastereo- and enantiocontrol to prepare a chiral skeleton bearing two stereocentres has received research interest (Fig. 1b). Recently, stereodivergent synthesis of a molecule featuring both enantioselectivity and *Z*/*E* selectivity has also been reported by several elegant works[22–26] (Fig. 1b). In particular, Zhang and coworkers described an important breakthrough in the stereodivergent synthesis of non-natural amino acids through precise control of enantio- and geometric selectivity, highlighting both the value and challenges of two-dimensional divergent synthesis[24]. By contrast, simultaneous control of more stereoselectivity

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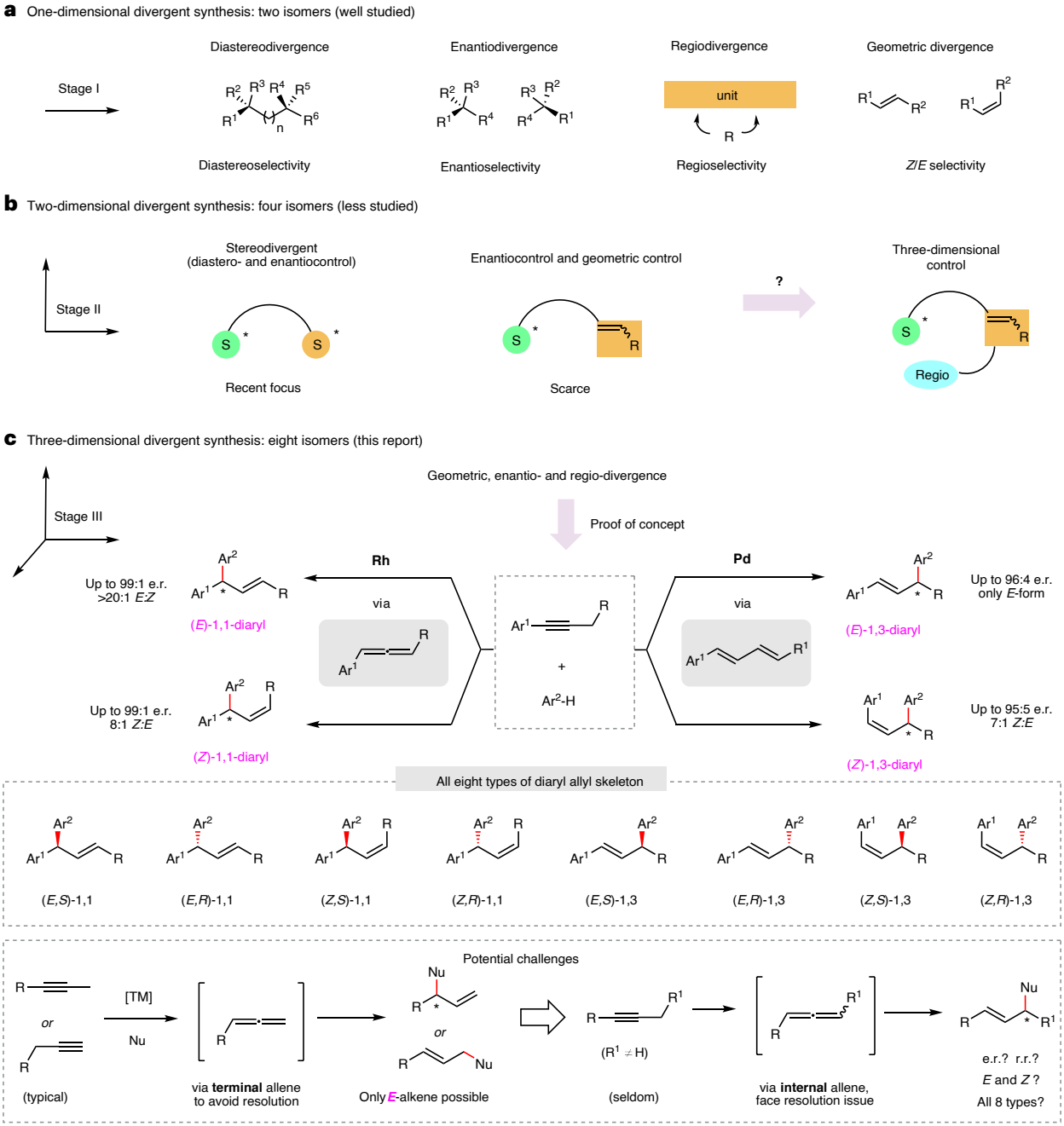


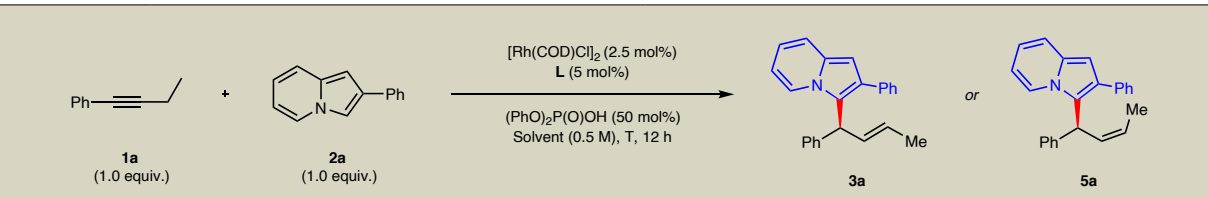
Fig. 1 | Different stages of divergent synthesis. a, One-dimensional divergent synthesis. **b**, Two-dimensional divergent synthesis. **c**, This design: three-dimensional divergent synthesis. Nu, nucleophile; Ar, aryl; TM, transition metal.

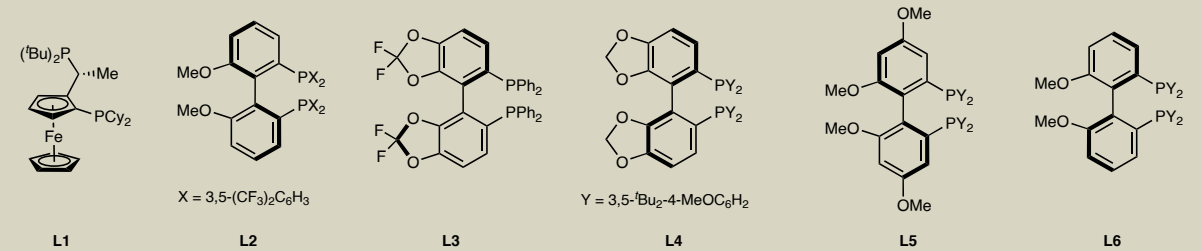
factors, such as three-dimensional divergent synthesis, is currently underexplored.

Alkynes are feedstock chemicals and widely used in organic synthesis. The transformation of simple alkynes into complex and valuable chiral building blocks is receiving much attention. Among them, transition-metal-catalysed asymmetric hydrofunctionalization of alkynes has emerged as a powerful route to construct stereocentres, owing to high stereoselectivity, readily available starting materials, mild reaction conditions and 100% atom economy of the process[27–30]. Based on the remarkable contributions from Yamamoto, Breit, Dong, Gong and many others, mechanistically, it is generally accepted that in the hydrofunctionalization the alkyne moiety must first be transformed into an allene unit[27–32]. Then, the allene intermediate receives sequential hydrometallation and allylic substitution to complete the reaction. For improved reactivity, terminal alkynes and propyne derivatives are typically adopted as substrates for hydrofunctionalization[33–50] (Fig. 1c, bottom). In this case, only a terminal allene intermediate is formed. However, stereoselective hydrofunctionalization of internal alkynes leading to internal allene intermediates has been seldom studied (Fig. 1c, bottom). This may be because the same chiral metal catalyst must not only ensure the formation of an internal allene intermediate with the correct configuration but also exert high stereocontrol over the subsequent hydrofunctionalization step. If a racemic allene or allene intermediate in low enantioselectivity was formed, the following hydrofunctionalization would require a challenging resolution process. Meanwhile, the construction of thermodynamically less stable (*Z*)-alkene or even *Z*- and *E*-controllable allyl compounds via asymmetric hydrofunctionalization of alkynes is also unknown.

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Table 1 | Reaction development for stereodivergent hydroheteroarylation of alkynes





Entry	L	Solvent	Temperature (°C)	Yield (%)[a]	*Z*/*E* (5a:3a)[a]	e.r.[b]
1	**L1–L3**	2-MeTHF	80	n.d.		
2	**L4**	2-MeTHF	80	10		96:4 (*E*)
3	**L5**	2-MeTHF	80	10		95:5 (*E*)
4	**L6**	2-MeTHF	80	12		96:4 (*E*)
5	**L6**	DCE, toluene, MeOH, CPME, dioxane, CH_2Cl_2	80	Trace		
6	**L6**	DMF	30	24	1:35	98:2 (*E*)
7	**L6**	DMF	50	49	1:11	98:2 (*E*)
8[c]	**L6**	DMF	50	67	1:12	98:2 (*E*)
9[c,d]	**L6**	DMF	50	72	1:12	99:1 (*E*)
10[c,d,e]	**L6**	DMF	50	78 (74)[f]	1:23 (1:14)[f]	99:1 (*E*)
11[c,d]	**L6**	2-MeTHF	50	32	1.5:1	98:2 (*Z*)
12[c,d]	**L5**	2-MeTHF	50	33	2.5:1	98:2 (*Z*)
13[c,d]	**L4**	2-MeTHF	50	46	2.6:1	99:1 (*Z*)
14[c,g]	**L4**	2-MeTHF	50	43	3.4:1	98:2 (*Z*)
15[c,g,h]	**L4**	2-MeTHF	50	40	5:1	98:2 (*Z*)
16[c,g,i]	**L4**	2-MeTHF	50	68	1.84:1	98:2 (*Z*)
17[c,g,j]	**L4**	2-MeTHF	50	51[f]	7.7:1[f]	98:2 (*Z*)

[a]Determined by crude ^{1}H NMR. For entries 11–17, the reaction time was 24 h. [b]Determined by high-performance liquid chromatography (HPLC) analysis. [c]1a (1.5 equiv.). [d]$LiBF_4$ (5 mol%) was added as an additive. [e]MOE (5.0 equiv.) was added as an additive. [f]Isolated yield and *Z*/*E* ratio were redetermined by pure corresponding ^{1}H NMR. [g]LiOTf ($LiSO_3CF_3$, 5 mol%) was added as an additive. [h]$(PhO)_2P(O)OH$ (80 mol%) was used. [i]$(PhO)P(O)(OH)_2$ (80 mol%) was used as acid additives instead. [i]$(PhO)_2P(O)OH$ (70 mol%) and $(PhO)P(O)(OH)_2$ (10 mol%) were used as acid additives instead. DCE, 1,2-dichloroethane; CPME, cyclopentyl methyl ether; n.d., not detected; ^{t}Bu, *tert*-butyl; Cy, cyclohexyl.

Based on our efforts on asymmetric hydrofunctionalizations of unsaturated bonds[23,47,51–56], we envisaged that precise dictation of catalysts (both metal and chiral ligand) and mechanistic pathways might provide a potential route for the unexplored three-dimensional divergent synthesis (Fig. 1c). It is known that different transition metals, such as Rh and Pd, could provide allylation products with different regioselectivities. However, it is elusive whether and how these two catalysts undergo efficient asymmetric hydrofunctionalizations of internal alkynes, especially when facing the resolution issue from internal allene intermediate. Moreover, stereoselective formation of the (*Z*)-alkene product during this process has no prior success because of the known unfavourable η^3-substitution process[57]. Here, we report a three-dimensional divergent synthesis system that enables regio-, enantio- and *Z*/*E*-controllable preparation of enantioenriched diaryl allyl moieties from simple alkynes and heteroarenes. Experimental and computational studies unveil the unexpected mechanistic differences of Pd and Rh catalysis for hydroarylation of internal alkynes.

Results and discussion

Recently, Zhang reported a synergistic bimetal-catalysed stereodivergent synthesis with precise enantio- and geometric control[24]. Here, we expected to provide a different route to reach the same stereodivergence goal by relying on single metal catalyst but with slight condition changes. Based on this hypothesis, we initiated the study by using alkyne 1a and indolizine 2a as the substrates, diphenyl phosphate as Brønsted acid additives, and 2-methyltetrahydrofuran (2-MeTHF) as the solvent at 80 °C under Rh catalysis (Table 1). A range of chiral bisphosphine ligands were examined first. With Josiphos ligand L1 and several biaryl ligands (L2–L3) used, no desired hydrofunctionalization process occurred (Table 1, entry 1). When a couple of bisphosphine ligands (L4–L6) bearing similar substituents were adopted, fortunately, low yield but high enantioselectivity were observed for expected (*E*)-product 3a (entries 2–4). With L6 as the chiral ligand, then a series of trials on solvent and temperature changes were evaluated (entries 5–7; see Supplementary Section 3.1 for more details). It showed that

N,*N*-dimethylformamide (DMF) as the solvent and a lower temperature were favourable to the transformation, generating 3a in 49% yield, 11:1 *E*:*Z* and 98:2 enantiomeric ratio (e.r.) (entry 7). By increasing the loading of substrate 1a, the yield of 3a was further elevated to 67% and with retained stereoselectivities (entry 8). Considering that lithium salt might work as an activator to promote the reaction[58,59], a catalytic amount of $LiBF_4$ was added as an additive and increased the reaction yield of 3a slightly (entry 9). Finally, the addition of 5 equivalents of 2-methoxyethanol (MOE) to the reaction was favourable to the yield and *E*:*Z* selectivity of 3a, furnishing product 3a in an optimal 74% isolated yield, 14:1 *E*:*Z* and 99:1 e.r. (entry 10)[60,61]

During the optimization process of Rh-catalysed hydroheteroarylation above, a reversed geometric selectivity with 2-MeTHF as the solvent was observed, which provided 5a as the major isomer (Table 1, entry 11). Although the *Z*/*E* selectivity was low in this case, it suggested the possibility to prepare *Z*-form allyl skeleton via hydrofunctionalization of alkynes. Previously, Breit described Rh-catalysed *Z*-selective hydrofunctionalization of allenes. However, this case required the allene substrate to be symmetric to avoid selectivity issues; otherwise, very low *Z*/*E* ratios were observed[62,63]. Their work indicated that it would be challenging to establish a *Z*-selective hydrofunctionalization reaction of alkynes, because the present work was proposed to undergo the formation of an unsymmetric allene intermediate. With the deduction in mind, we still continued the examination of various chiral ligands and lithium additives for the transformation (entries 12–14; see Supplementary Section 3.2 for more details). It showed that L4 as the chiral ligand and LiOTf as additives were beneficial to the reaction and afforded 5a in 43% yield, 3.4:1 *Z*:*E* and 98:2 e.r. (entry 14). Increasing the amount of Brønsted acid further elevated *Z*:*E* selectivity to 5:1 for 5a (entry 15). However, other types of P(V)-based Brønsted acid seemed unfavourable to the reaction (see Supplementary Section 3.2 for details), except that $(PhO)P(O)(OH)_2$ as acid additives formed 5a in a higher yield but with lower geometry selectivity (entry 16). Ultimately, the suitable combination of both $(PhO)_2P(O)(OH)$ and $(PhO)P(O)(OH)_2$ as acid additives provided *Z*-allyl skeleton 5a in an optimal 51% isolated yield, 7.7:1 *Z*:*E* and 98:2 e.r. (entry 17).

Having identified the suitable conditions for stereodivergent hydroheteroarylation reaction, the scope of Rh-catalysed synthesis of enantioenriched (*E*)-1,1-diaryl skeletons was first examined, and the results are summarized in Fig. 2. Aryl butyne bearing different substituents at *para*- and *meta*-positions of arene units showed good compatibility with the transformation, generating corresponding (*E*)-1,1-diaryls in generally >70% yield, >20:1 regioisomeric ratio (r.r.), >10:1 *E*:*Z* and >98:2 e.r. (3a–3g). Naphthyl and heteroaryl-derived alkynes were also suitable substrates for the reaction (3h and 3i). As indolizine unit is ubiquitous in bioactive molecules[64,65], the tolerance of indolizine derivatizations as heteroarene nucleophiles was tested next. Aryl-substituted indolizines containing various functional groups, including ether, CF_3, F, Br, CN and alkyl units, reacted with alkyne 1a smoothly (3j–3o and 3t). Other derivatives, such as multisubstituted aryl, naphthyl, thienyl and cyclopropyl-derived indolizines were also well tolerated with the transformation (3p–3s). The absolute configuration of 3r was elucidated to be (*S*)-3r via X-ray crystallography analyses. When alkyne bearing a longer alkyl chain was used as the substrate, the regio-, geometry- and enantioselective hydrofunctionalization process proceeded well, affording corresponding product 3u in a moderate yield but with high three-dimensional selectivity control. The present protocol for the preparation of (*E*)-1,1-diaryls could be applied to hydroheteroarylation of typical propynes. For example, aryl alkynes containing groups such as alkyl, aryl, F and CF_3 units, as electrophiles displayed good reactivity and stereocontrol for the transformation (4a–4f). Similarly, a series of aryl, heteroaryl, alkyl and ester derivatives in indolizine skeleton were suitable for the reaction, providing enantioenriched 1,1-diaryls building blocks (4m–4t) in good yields and stereoselectivities. When dialkyl-substituted internal alkyne such as hex-3-yne was used as the substrate, no hydrofunctionalization occurred, indicating the importance of aryl unit in the alkyne substrate for the transformation (see Supplementary Fig. 1 for details).

Next, the reliability of *Z*-selective hydrofunctionalization of alkynes via Rh catalysis was assessed (Fig. 3a). A prior case from Breit showed that *E*-form hydrofunctionalization product was preferred when unsymmetric allene substrate was used, and γ-regioselectivity was observed[62]. By contrast, in our case, the hydrofunctionalization of alkynes via unsymmetric allene intermediate afforded (*Z*)-1,1-diaryls with generally >5:1 geometric selectivity and exclusive α-regioselectivity (5a–5g). A series of indolizine derivatives displayed good compatibility with the transformation. For example, naphthyl-derived indolizine reacted with aryl butyne 1a smoothly, furnishing corresponding product 5c in 55% yield, >20:1 r.r., 8:1 *Z*/*E* and 98:2 e.r.

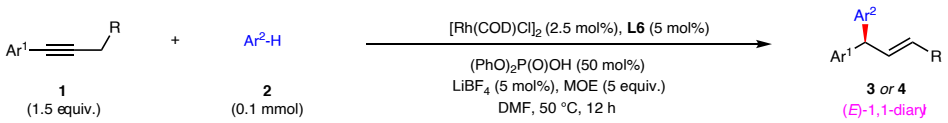
In addition to Rh-catalysed stereodivergent hydrofunctionalization of alkynes, a Pd-catalysed process was also explored to achieve third-dimensional control over regioselectivity. With $[Pd(allyl)Cl]_2$ and L1 as the catalyst, Et_3N as the base, PhCOOH and *n*-PrOH as additives and tetrahydrofuran (THF) as the solvent, the reaction was conducted with the same substrates 1a and 2a as used in the Rh-catalysed process and delivered (*E*)-1,3-diaryl skeleton 6a in 76% yield, >20:1 r.r., >20:1 *E*/*Z* and 96:4 e.r. (Fig. 3b). A variety of alkynes bearing diverse substituents, such as alkyl, phenyl, F, CF_3, CF_3O and thiazolyl, were well tolerated in the transformation, affording the corresponding products in good yields with high regio-, geometric and enantioselectivities (6b–6k). Indolizine derivatives as nucleophiles exhibited similarly high functional group tolerance. Different aryl, hetereoaryl, alkyl and small ring units at varying positions of indolizine skeleton were suitable for the stereoselective hydroheteroarylation (6l–6y). Another alkyl alkyne also underwent the reaction in moderate yields and with excellent stereoselectivities (6z). The absolute configuration of 6p was elucidated to be (*R*)-6p via X-ray crystallography analyses. However, when a dialkyl-substituted internal alkyne, such as hex-3-yne, was used as the substrate, hydrofunctionalization proceeded with low reactivity, with substitution occurring mainly at the terminal position. In addition, other aryl carbon nucleophiles were evaluated but showed no reactivity under either Pd or Rh catalysis under the current conditions, highlighting the challenge of hydrofunctionalizing internal alkynes (see Supplementary Fig. 1 for details).

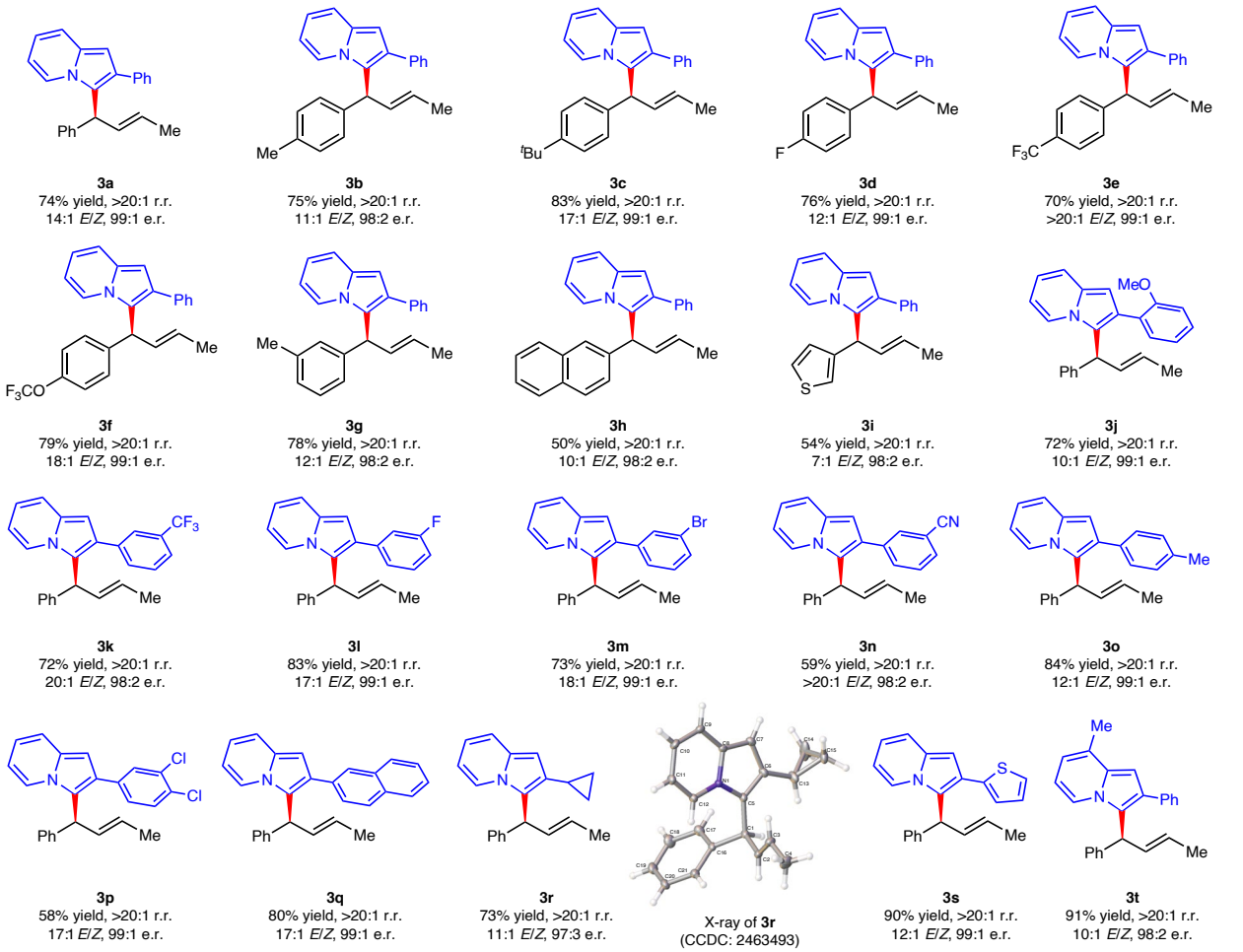
As (*E*)-alkene might isomerize to (*Z*)-alkene under suitable photocatalysts[14,26,66], we envisioned that enantioenriched (*Z*)-1,3-diaryls might be prepared through hydroheteroarylation of alkynes via Pd/Ir relay catalysis. Under this design, a series of enantioenriched (*Z*)-1,3-diaryl skeletons were prepared from the reaction of alkynes 1 and indolizines 2 in generally >20:1 r.r., >4:1 *Z*/*E* and >94:6 e.r. (Fig. 3c, 7a–7f). For example, a (*Z*)-1,3-diaryl scaffold 7f was synthesized conveniently through this cascade process in 70% yield, >20:1 r.r., 7:1 *Z*/*E* and 94:6 e.r. At present, the presence of an *ortho*-substituent on the aryl alkyne substrate 1 is required to ensure good geometric selectivity in the transformation (7g and 7h).

To highlight the power of the present synthetic protocol, three-dimensional divergent synthesis was conducted (Fig. 4a). With simple alkyne 1a or analogue and heteroarene 2a as the starting materials, all eight types of related chiral diaryl allyl skeleton with dictation at geometric, regio- and enantioselectivity were prepared smoothly. This divergent synthetic strategy demonstrates the possibility to achieve higher-dimensional selectivity control in molecule design, which is a long-standing challenge but important goal in synthetic chemistry area. In addition, the reliability of the protocol was corroborated by scale-up tests. Both 3a and 6a were obtained in increased yields and maintained selectivities (Fig. 4b). Further hydrogenation smoothly afforded the diaryl alkyl analogues 3aa and 6aa.

To understand the mechanistic differences in Pd- and Rh-catalysed hydrofunctionalization of alkynes, a series of experiments was

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3a
74% yield, >20:1 r.r.
14:1 *E*/*Z*, 99:1 e.r.

3b
75% yield, >20:1 r.r.
11:1 *E*/*Z*, 98:2 e.r.

3c
83% yield, >20:1 r.r.
17:1 *E*/*Z*, 99:1 e.r.

3d
76% yield, >20:1 r.r.
12:1 *E*/*Z*, 99:1 e.r.

3e
70% yield, >20:1 r.r.
>20:1 *E*/*Z*, 99:1 e.r.

3f
79% yield, >20:1 r.r.
18:1 *E*/*Z*, 99:1 e.r.

3g
78% yield, >20:1 r.r.
12:1 *E*/*Z*, 98:2 e.r.

3h
50% yield, >20:1 r.r.
10:1 *E*/*Z*, 98:2 e.r.

3i
54% yield, >20:1 r.r.
7:1 *E*/*Z*, 98:2 e.r.

3j
72% yield, >20:1 r.r.
10:1 *E*/*Z*, 99:1 e.r.

3k
72% yield, >20:1 r.r.
20:1 *E*/*Z*, 98:2 e.r.

3l
83% yield, >20:1 r.r.
17:1 *E*/*Z*, 99:1 e.r.

3m
73% yield, >20:1 r.r.
18:1 *E*/*Z*, 99:1 e.r.

3n
59% yield, >20:1 r.r.
>20:1 *E*/*Z*, 98:2 e.r.

3o
84% yield, >20:1 r.r.
12:1 *E*/*Z*, 99:1 e.r.

3p
58% yield, >20:1 r.r.
17:1 *E*/*Z*, 99:1 e.r.

3q
80% yield, >20:1 r.r.
17:1 *E*/*Z*, 99:1 e.r.

3r
73% yield, >20:1 r.r.
11:1 *E*/*Z*, 97:3 e.r.

X-ray of **3r**
(CCDC: 2463493)

3s
90% yield, >20:1 r.r.
12:1 *E*/*Z*, 99:1 e.r.

3t
91% yield, >20:1 r.r.
10:1 *E*/*Z*, 98:2 e.r.

4a, Ar^1 = Ph, 74% yield, >20:1 r.r., 95:5 e.r.
4b, Ar^1 = *p*-MeC_6H_4, 88% yield, >20:1 r.r., 94:6 e.r.
4c, Ar^1 = *p*-$^tBuC_6H_4$, 83% yield, >20:1 r.r., 95:5 e.r.[b]
4d, Ar^1 = (*p*-Ph)C_6H_4, 75% yield, >20:1 r.r., 93:7 e.r.
4e, Ar^1 = *p*-FC_6H_4, 76% yield, >20:1 r.r., 94:6 e.r.[b]
4f, Ar^1 = (*p*-CF_3)C_6H_4, 71% yield, >20:1 r.r., 91:9 e.r.[b]
4g, Ar^1 = *p*-ClC_6H_4, 78% yield, >20:1 r.r., 93:7 e.r.[b]
4h, Ar^1 = (*m*-CF_3)C_6H_4, 77% yield, >20:1 r.r., 91:9 e.r.
4i, Ar^1 = *m*-MeC_6H_4, 88% yield, >20:1 r.r., 92:8 e.r.
4j, Ar^1 = 2-naphthyl, 82% yield, >20:1 r.r., 95:5 e.r.
4k, Ar^1 = 1-naphthyl, 48% yield, >20:1 r.r., 96:4 e.r.
4l, Ar^1 = 3-thiazolyl, 58% yield, >20:1 r.r., 95:5 e.r.[b]

3u
51% yield, >20:1 r.r.
15:1 *E*/*Z*, 98:2 e.r.[a]

4m, Ar = (*o*-MeO)C_6H_4, 92% yield, >20:1 r.r., 95:5 e.r.
4n, Ar = (*m*-MeO)C_6H_4, 80% yield, >20:1 r.r., 94:6 e.r.
4o, Ar = *m*-FC_6H_4, 66% yield, >20:1 r.r., 94:6 e.r.
4p, Ar = (*m*-CF_3)C_6H_4, 87% yield, >20:1 r.r., 95:5 e.r.
4q, Ar = *p*-MeC_6H_4, 88% yield, >20:1 r.r., 92:8 e.r.
4r, Ar^1 = 2-thiazolyl, 96% yield, >20:1 r.r., 92:8 e.r.

4s, 79% yield
>20:1 r.r., 94:6 e.r.

4t, 68% yield
5:1 r.r., 93:7 e.r.[c]

Y = 3,5-$^tBu_2$-4-$MeOC_6H_2$
L6a

Fig. 2 | Rh-catalysed hydroheteroarylation of alkynes to prepare (*E*)-1,1-diaryls. Isolated yield. The e.r. was determined by HPLC analysis. The r.r. and *E*/*Z* ratios were determined by ^1H NMR analysis. For the preparation of **4**, the following reaction conditions were adopted: **1** (1 equiv.), **2** (2 equiv.), $[Rh(COD)Cl]_2$ (2.5 mol%), **L5** (5 mol%), $(BnO)_2P(O)OH$ (10 mol%) in 2-MeTHF at 80 °C for 24 h. [a]$[Rh(COD)Cl]_2$ (5 mol%), **L6** (10 mol%) and $LiBF_4$ (10 mol%) were used. [b]**L6a** was used as the chiral ligand. [c]$(PhO)_2P(O)OH$ (50 mol%) at 60 °C for 3 h was adopted.

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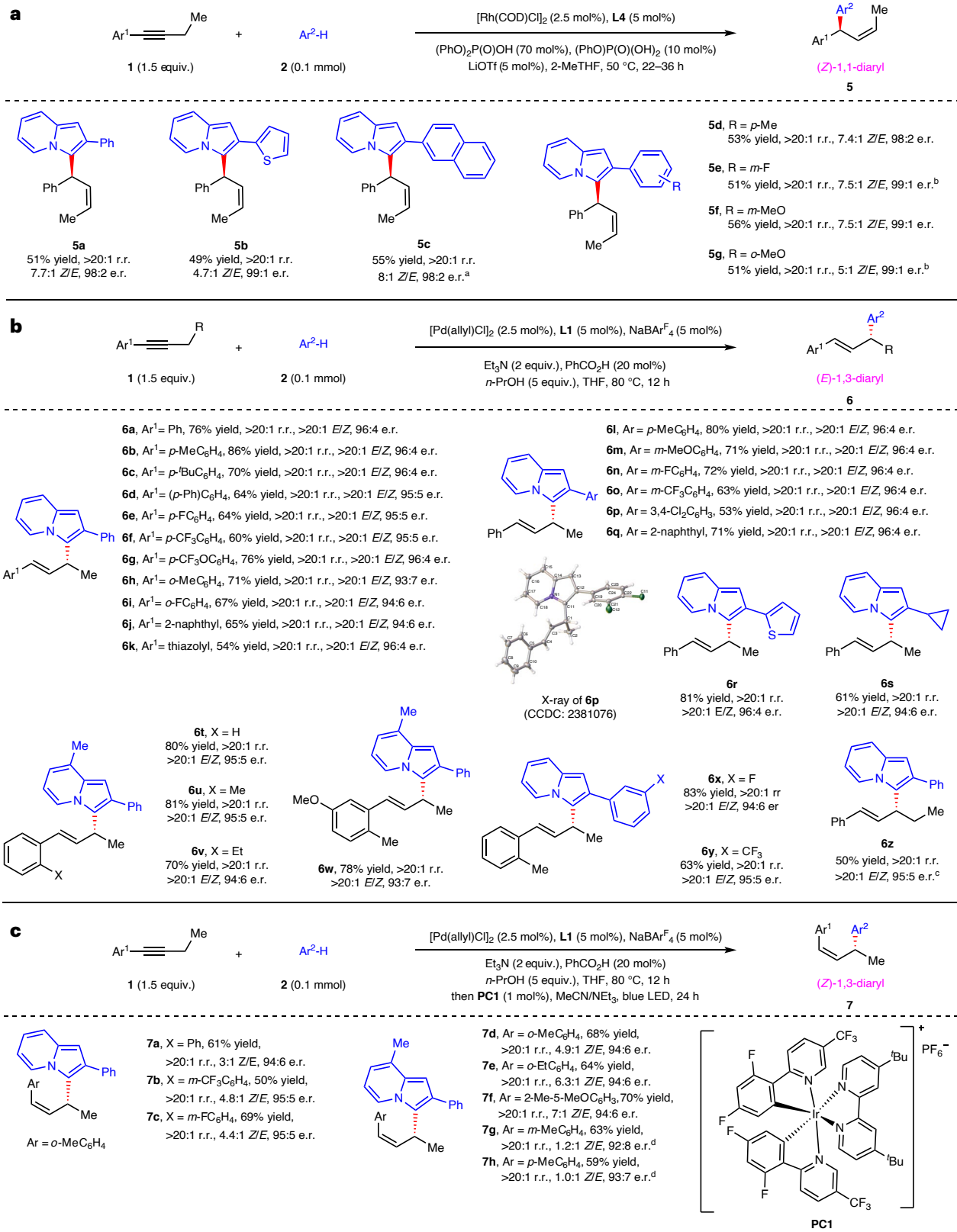


Fig. 3 | Divergent hydroheteroarylation of alkynes. a, Rh-catalysed synthesis of (*Z*)-1,1-diaryls. **b**, Pd-catalysed synthesis of (*E*)-1,3-diaryls. **c**, Pd-catalysed synthesis of (*Z*)-1,3-diaryls. Isolated yield as the mixture of *E*/*Z* isomers. The e.r. was determined by HPLC analysis. The r.r. and *E*/*Z* ratios were determined by ^{1}H NMR analysis. [a](PhO)$_2$P(O)OH (80 mol%) was used as acid additives. [b][Rh(COD)Cl]$_2$ (5.0 mol%), L4 (10 mol%) and (PhO)$_2$P(O)OH (80 mol%) were adopted. [c][Pd(allyl)Cl]$_2$ (5 mol%), L1 (10 mol%), NaBAr$^F_4$ (10 mol%), PhCO$_2$H (50 mol%) and MeCN (solvent) were used instead. [d]Alkyne **1** (1.2 equiv.) was used. [n]Pr, *n*-propyl; Ar^F, 3,5-(CF$_3$)$_2$C$_6$H$_3$; blue LED, blue-light-emitting diode (455 nm, 10 W).

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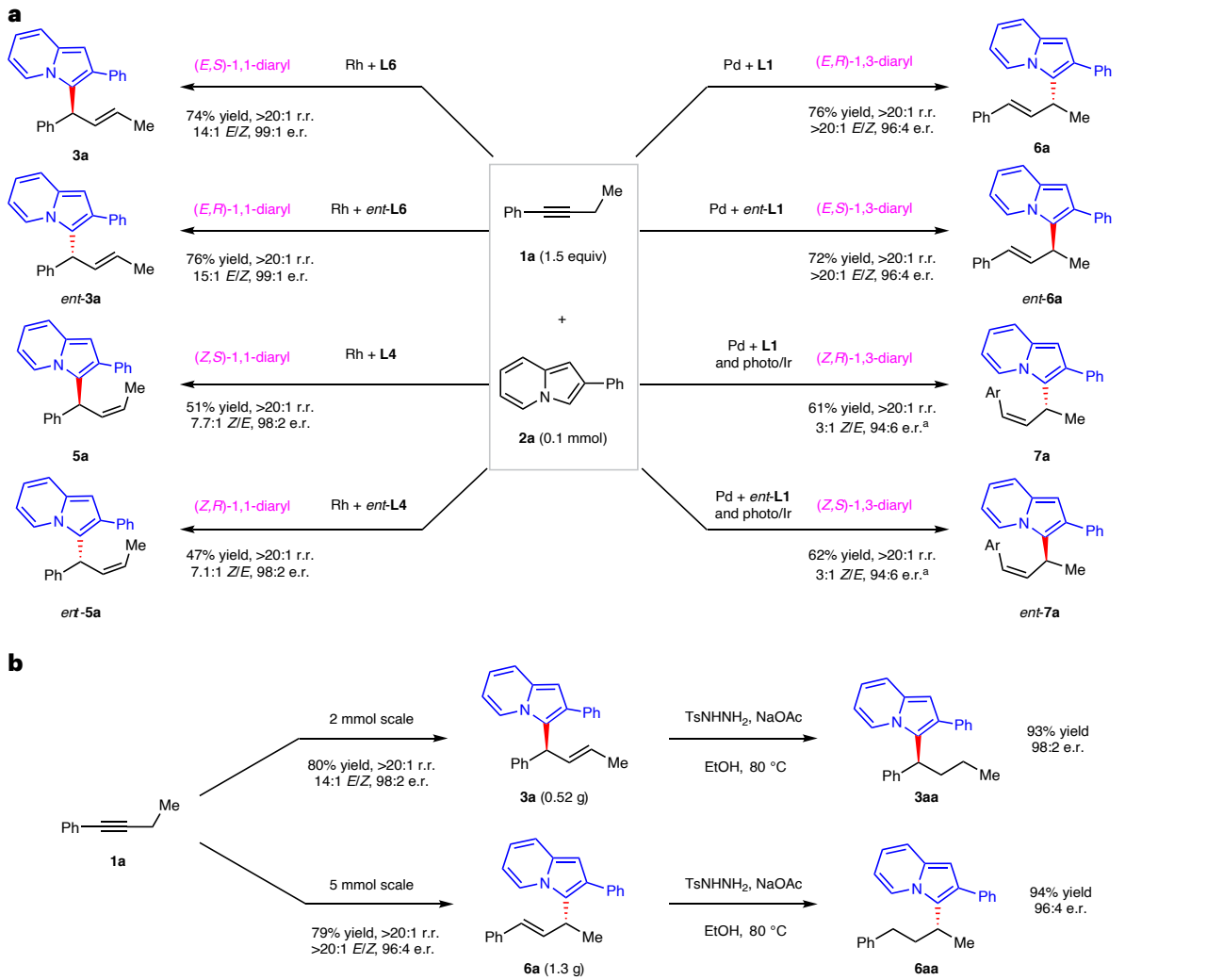


Fig. 4 | Three-dimensional divergent synthesis, gram-scale test and transformations. a, Three-dimensional divergent synthesis. **b**, Gram-scale test and transformations. [a]Ar, *o*-MeC$_6$H$_4$; TsNHNH$_2$, 4-methylbenzenesulfonhydrazide; Ac, acetyl.

conducted (Fig. 5). With deuterated 1b as the substrates, 3b-*d* was obtained smoothly in similarly high stereoselectivity via Rh catalysis. In this case, the terminal methyl unit in 3b-*d* was observed with 100% retained deuteration without any erosion (Fig. 5a, left). This result suggested that no transformation occurred at the terminal methyl site. By contrast, when the same reaction was conducted under Pd catalysis, a decrease in deuteration was observed at the terminal methyl unit in product 6b-*d* (Fig. 5a, right). This showed that a different transformation process might occur with Pd catalyst, which presumably involved a β-H elimination step to cleave the terminal deuterium atom[23]. This proposal was further corroborated by another set of deuterium-labelling experiments by using deuterated acid and alcohol additives for the transformations (see Supplementary Section 9.7 for details). Next, two control experiments were conducted to uncover the potential intermediates (Fig. 5b). With alkyne 8 as the substrate under Rh catalyst in 1 h reaction time, the corresponding product 3u was detected in trace amounts, along with trace amounts of conjugated diene 9 and no allene intermediate 10 (Fig. 5b, left). We thought that the formed allene species might be reactive and quickly consumed during the reaction. However, 15% conjugated diene intermediate 9 was detected under Pd catalyst (Fig. 5b, right). Based on these results, we deduced that the Pd-catalysed hydroarylation of alkynes probably proceeds through the formation of diene intermediates, consistent with the observed decrease in deuteration shown in Fig. 5a. To corroborate the hypothesis, diene 11 was directly used as the substrate instead of alkyne. It showed no reactivity towards Rh catalyst. By comparison, good yield and stereoselectivity were observed for 6a when using diene 11 as the substrate under Pd catalyst (Fig. 5c). These tests further support the proposal that hydroarylation of internal alkynes proceeds via sequential formation of allene and diene intermediates under Pd catalysis, whereas only allene intermediate formation occurs under Rh catalysis.

The formation of an allene or diene intermediate could further be determined by using a special alkyne 12 as the substrate (Fig. 5d). With Rh catalyst, only in situ hydrofunctionalization product 13 was obtained (Fig. 5d, left). This result provided additional support for a transformation process involving sequential allene formation, hydrometallation and allylic substitution steps for the present Rh-catalysed hydroarylation reaction. By contrast, product 6a was detected, instead of a terminal substitution product, through in situ hydrofunctionalization via Pd-catalysed hydroarylation of alkyne 12 (Fig. 5d, right). The observation of comparable reactivity and stereoselectivity using different substrates—alkyne 1a, diene 11 and alkyne 12—under Pd catalysis further demonstrates the facile isomerization among these three species. Similarly, direct adoption of terminal allene 12′ as the substrate exhibited the same trend, which served as additional support to the aforementioned conclusion (Fig. 5d, bottom). Finally, when racemic allene 14 was used as the substrate, both products 3a and 5a were observed under Rh catalyst (Fig. 5e, left). Similarly, racemic allene 14 was also a reactive

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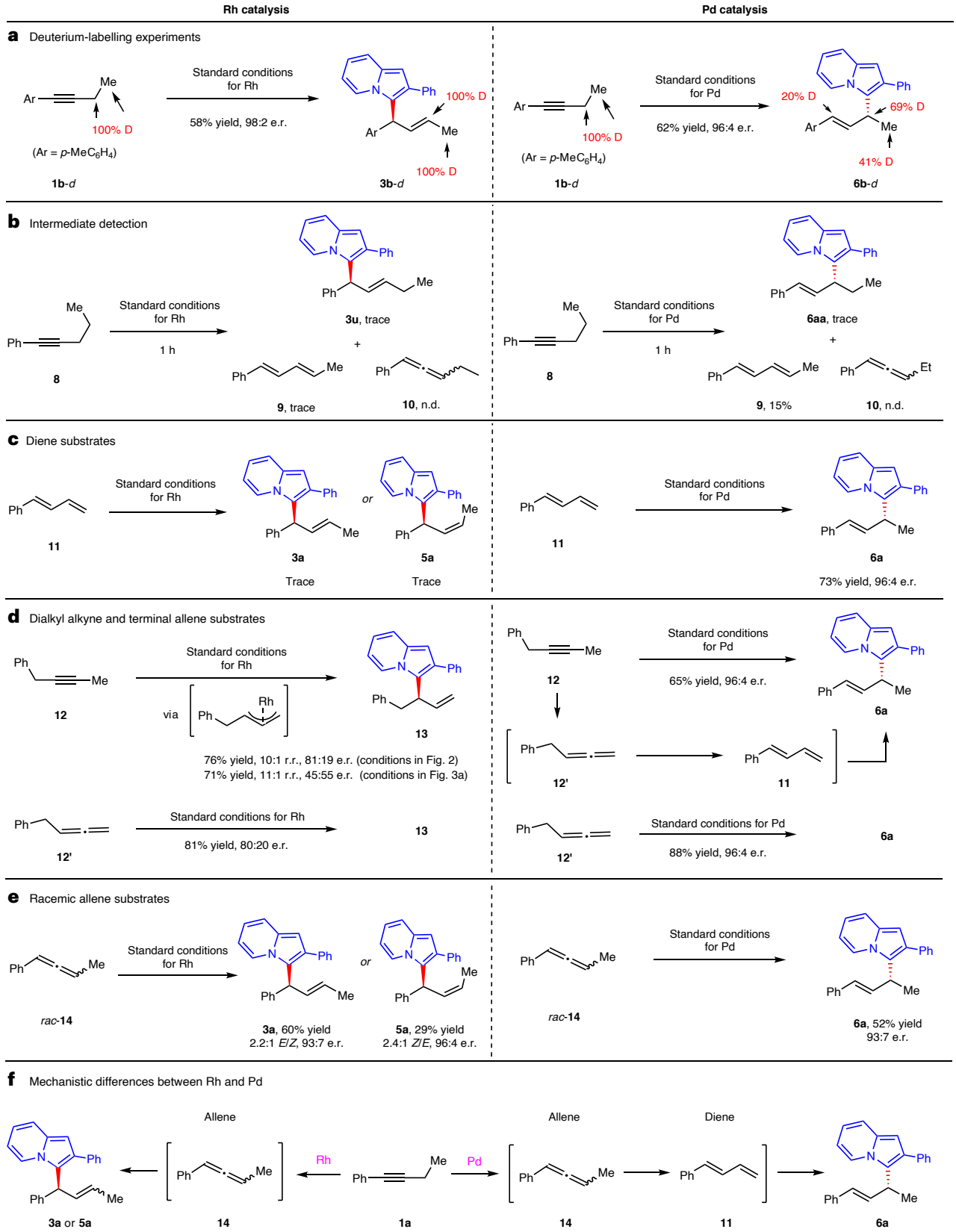


Fig. 5 | Mechanistic elucidation of Rh- and Pd-catalysed hydrofunctionalizations of internal alkynes. a, Deuterium-labelling experiments. **b**, Intermediate detection. **c**, Diene as the substrate for the reaction. **d**, Dialkyl alkyne as the substrate. **e**, Racemic allene as the substrate. **f**, Mechanistic differences between Rh- and Pd-catalysed processes.

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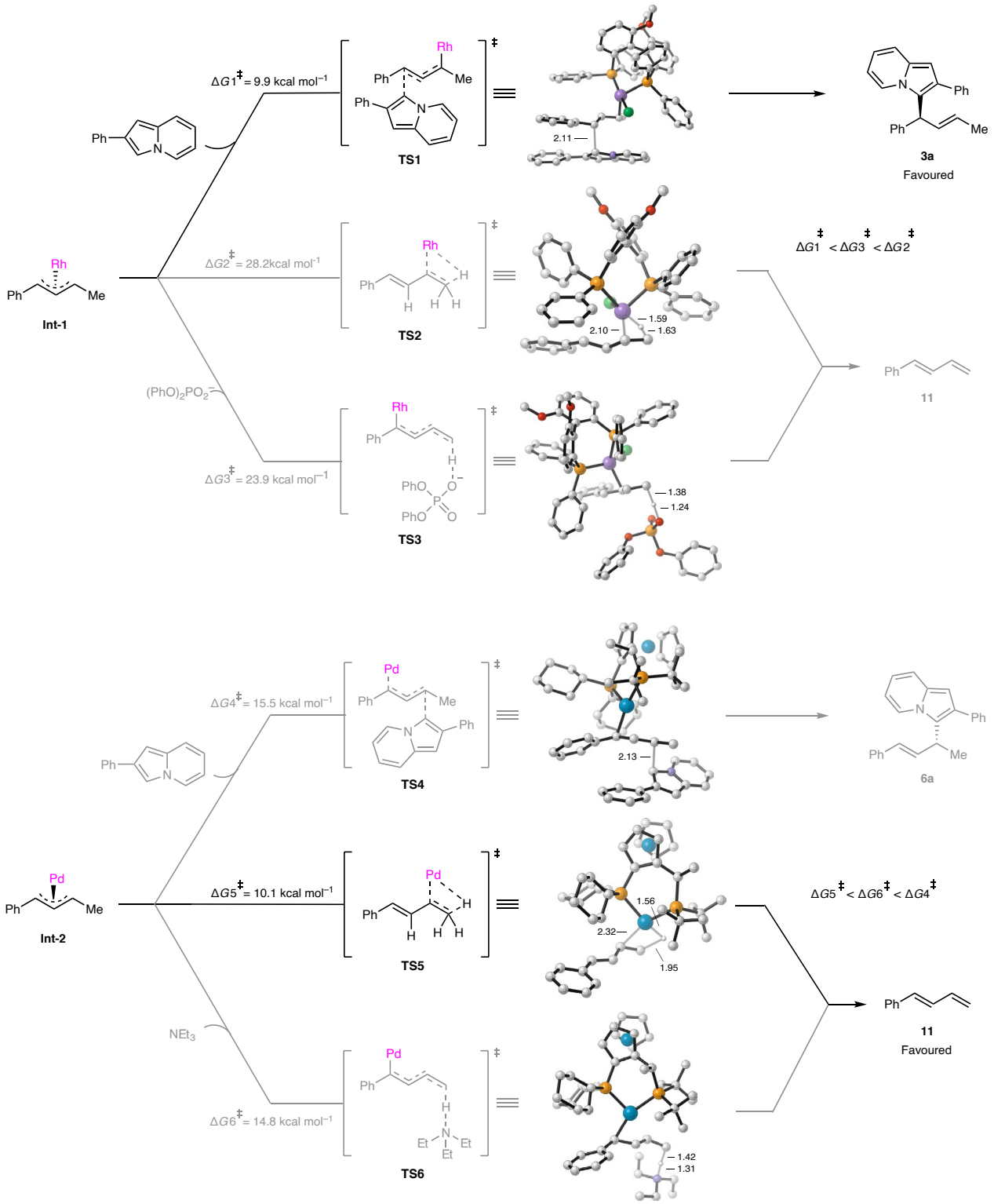


Fig. 6 | DFT calculations on the origin of different mechanistic pathways. For the Rh catalytic route, L10 as the simplified ligand of L6 was used as the ligand to reduce the amount of calculation (see Supplementary Section 10 for details). For the Pd catalytic route, the standard ligand L1 was used.

substrate for Pd-catalysed hydroarylation. These results are consistent with the widely accepted view that allene species serve as intermediates in transition-metal-catalysed hydrofunctionalization of alkynes.

Based on all these mechanistic experiments, we can conclude that Rh and Pd catalysts undergo different mechanistic pathways for stereoselective hydroheteroarylation of internal alkynes (Fig. 5f). For Rh-catalysed reaction, a typical isomerization of the alkyne to an allene occurs. In this case, the enantioselectivity of the formed allene intermediate might affect the stereocontrol of the final allylic substitution product. However, a Pd-catalysed process could avoid this potential issue, because it can transform the alkyne into an allene and then generate a conjugated diene intermediate.

To further elucidate the origin of the different mechanistic pathways, we conducted density functional theory (DFT) calculations focusing on the key π-allyl complexes (Fig. 6). Computational results reveal that the π-allyl rhodium complex (Int-1) preferentially

undergoes allylic substitution with an energetically favourable barrier of 9.9 kcal mol^{-1} ($\Delta G1^{‡}$). By contrast, β-H elimination exhibits a much higher barrier of 28.2 kcal mol^{-1} ($\Delta G2^{‡}$), indicating the low reactivity of the Rh catalyst towards β-H elimination. Further consideration was given to the role of the phosphate anion in the reaction system. Although the participation of the phosphate anion slightly promotes β-H elimination ($\Delta G3^{‡}$ = 23.9 kcal mol^{-1}), allylic substitution remains the dominant pathway due to its lower energy barrier, as dictated by thermodynamics. The sequence of energy barriers ($\Delta G1^{‡} < \Delta G3^{‡} < \Delta G2^{‡}$) suggests that 3a would be the main product, consistent with experimental observations. In contrast to the π-allyl rhodium complex (Int-1), the π-allyl palladium complex (Int-2) exhibits a preference for β-H elimination ($\Delta G5^{‡}$ = 10.1 kcal mol^{-1}) over allylic substitution ($\Delta G4^{‡}$ = 15.5 kcal mol^{-1}). Additional calculations on β-H elimination in the presence of the organic base NEt_3 reveal a less favourable step ($\Delta G6^{‡}$ = 14.8 kcal mol^{-1}), indicating that NEt_3 does not provide effective assistance to this pathway. The sequence of energy barriers for the Pd-catalysed reaction is $\Delta G5^{‡} < \Delta G6^{‡} < \Delta G4^{‡}$, which accounts for the formation of diene 11. These DFT calculations, together with the mechanistic experiments, provide a detailed understanding of the distinct catalytic mechanisms of Rh and Pd catalysts, explaining the observed product distributions in the experiments.

Conclusion

As a proof of concept, a three-dimensional divergent synthesis protocol is established via Pd and Rh catalysts. All eight types of diaryl allyl skeleton ((*E*,*R*)-1,1; (*E*,*S*)-1,1; (*E*,*R*)-1,3; (*E*,*S*)-1,3; (*Z*,*R*)-1,1; (*Z*,*S*)-1,1; (*Z*,*R*)-1,3; (*Z*,*S*)-1,3) are conveniently synthesized with good geometric, regio- and enantiocontrol from hydroheteroarylation reaction of simple alkynes and heteroarenes. Stereodivergent preparation of enantioenriched 1,1-diaryls bearing *Z*- or *E*-alkene moieties is achieved in high geometric, regio- and enantioselectivity via Rh catalyst ligated to different chiral ligands. Similarly, stereodivergent preparation of enantioenriched 1,3-diaryls bearing *Z*- or *E*-alkene moieties is achieved via Pd or Pd/Ir relay catalysis. Mechanistic experiments and DFT calculations reveal that Pd and Rh catalysts follow distinct catalytic pathways for the hydroarylation of alkynes, particularly differing in the deracemization mechanisms of the less-studied internal allene intermediate.

Methods

General procedure for Rh-catalysed hydroheteroarylation to prepare (*E*)-1,1-diaryls

In a nitrogen-filled glovebox, $[Rh(COD)Cl]_2$ (1.2 mg, 0.0025 mmol, where COD is 1,5-cyclooctadiene), L6 (5.8 mg, 0.0050 mmol), $(PhO)_2P(O)OH$ (13 mg, 0.050 mmol), $LiBF_4$ (0.5 mg, 0.0050 mmol), heteroarene 2 (0.10 mmol) and dry DMF (0.20 ml) were added sequentially to a 4-ml vial. Then, the reaction was stirred at room temperature for 1 min. Next, MOE (39 µl, 0.50 mmol) and alkyne 1 (0.15 mmol) were added to the reaction. The resulting mixture was stirred at 50 °C for 12 h. Afterwards, the reaction was concentrated and purified by flash column chromatography (*n*-hexane/EtOAc = 30:1) to afford the pure product 3.

General procedure for Rh-catalysed hydroheteroarylation to prepare (*Z*)-1,1-diaryls

In a nitrogen-filled glovebox, $[Rh(COD)Cl]_2$ (1.2 mg, 0.0025 mmol), L4 (5.9 mg, 0.0050 mmol), $(PhO)_2P(O)OH$ (18 mg, 0.070 mmol), $(PhO)P(O)(OH)_2$ (1.7 mg, 0.010 mmol), LiOTf (0.8 mg, 0.0050 mmol), heteroarene 2 (0.10 mmol) and dry 2-MeTHF (0.20 ml) were added sequentially to a 4-ml vial. Then, the reaction was stirred at room temperature for 1 min. Next, alkyne 1 (0.15 mmol) was added to the reaction and the resulting mixture was stirred at 50 °C for 24 h. Afterwards, the reaction was concentrated and purified by flash column chromatography to afford the pure product 5.

General procedure for Pd-catalysed hydroheteroarylation to prepare (*E*)-1,3-diaryls

In a nitrogen-filled glovebox, $[Pd(allyl)Cl]_2$ (0.9 mg, 0.0025 mmol), L1 (2.8 mg, 0.0050 mmol), $NaBAr^F_4$ (4.4 mg, 0.0050 mmol), $PhCO_2H$ (2.5 mg, 0.020 mmol), heteroarene 2 (0.10 mmol) and dry THF (0.20 ml) were added sequentially to a 4-ml vial. Then, the reaction was stirred at room temperature for 1 min. Next, *n*-propanol (37 µl, 0.50 mmol), Et_3N (28 µl, 0.20 mmol) and alkyne 1 (0.15 mmol) were added to the reaction, and the resulting mixture was stirred at 80 °C for 12 h. Afterwards, the reaction was concentrated and purified by flash column chromatography (*n*-hexane/EtOAc = 30:1) to afford the pure product 6.

General procedure for Pd-catalysed hydroheteroarylation to prepare (*Z*)-1,3-diaryls

In a nitrogen-filled glovebox, $[Pd(allyl)Cl]_2$ (0.9 mg, 0.0025 mmol), L1 (2.8 mg, 0.0050 mmol), $NaBAr^F_4$ (4.4 mg, 0.0050 mmol), $PhCO_2H$ (2.5 mg, 0.020 mmol), heteroarene 2 (0.10 mmol) and dry THF (0.20 ml) were added sequentially to a 4-ml vial. Then, the reaction was stirred at room temperature for 1 min. Next, *n*-propanol (37 µl, 0.50 mmol), Et_3N (28 µL, 0.20 mmol) and alkyne 1 (0.15 mmol) were added to the reaction and the resulting mixture was stirred at 80 °C for 12 h. Afterwards, the reaction mixture was filtered by silica gel with hexane (200 ml) as elute. The elute was collected and concentrated. Next, PC1 ($Ir[dF(CF_3)ppy]_2(dtbbpy)PF_6$, 1.2 mg, 0.0010 mmol), MeCN (0.80 ml) and Et_3N (0.20 ml) were added. The resulting new mixture was irradiated under blue-light-emitting diode (455 nm, 10 W) for 24 h. Afterwards, the reaction was concentrated and purified by flash column chromatography (*n*-hexane/EtOAc = 30:1) to afford the pure product 7.

Data availability

All relevant data are available with the Article and its Supplementary Information. Crystallographic data for the structures reported in this Article have been deposited at the Cambridge Crystallographic Data Centre, under deposition numbers CCDC 2463493 (3r) and 2381076 (6p). Copies of the data can be obtained free of charge at https://www.ccdc.cam.ac.uk/structures/.

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Author contributions

Z.-T.H. conceived the project. A.H. and Z.-J.Y. performed the experiments. H.-R.X. and X.-S.X. conducted the computations. Z.-T.H. and X.-S.X. supervised the project and wrote the manuscript with the feedback from all authors.

Competing interests

The authors declare no competing interests.

Additional information

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